IP-A1 Series
Remote API (HTTP)
Version 2
Specification

Rev.2.4.0

2025/08/27

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■Revision 2.5History

Rev	Date	Content
2.0.0	2022/7/21	Initial Version
2.1.0	2022/10/20	API v2.1
2.1.1	2023/02/24	Add description to section 2.5 on HTTP Status code 401
	2023/02/24	Remove duplicate page.
	2023/02/24	Fix wrong description on when the setting will be applied at section 10.6
2.2.0	2023/09/06	Support API v2.2 See 1.5.3
		Update firmware version table. See 1.5.1
2.3.0	2024/02/292 024/02/09	Add description on character encoding (See 1.4)
		Add error codes (See 2.5)
		Add media and multicast category (See chapter 4 , See 9.1 , See 9.2.12, See 9.2.13, See 10.12, See 10.13)
		Add examples to chapter 11
2.4.0	2024/08/27	Add support for https (See 1.2, 1.5.1 chapter 12)

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1 Description

This document describes remote API(HTTP) v2 specification.

1.1 Feature

This remote API supports the following features.

Supported commands are different on each model.

- Authentication
- Control system
- Update firmware
- Upload/download config file including internal audio files
- Start/stop SIP broadcasting
- Start/stop internal audio file
- Get information of internal audio
- Get/set audio volume
- Mute/unmute/get the setting
- Get operating status
- Start/stop local broadcasting
- Start/stop multicast transmission from Pattern or LINE/MIC input
- Send command sets
- Get control-in/out state, set control-out status
- Change external amplifier state
- Start/stop microphone broadcasting
- Start/stop recorded audio which is assigned to a pattern
- Set/Get receiving multicast channel
- Upload internal audio file

1.2 Protocol

This remote API is based on HTTP/1.1.

It supports http and https.

1.3 Protocol Version

This specification defines remote API version 2 (Ver 2.3) (See Chapter 2)

The API extends remote API version 1 keeping upward compatibility. (See Chapter 3)

Firmware which is compatible with this specification supports remote API version 1 unless referring at Chapter 3 explicitly.

1.4 Character Encoding

All members are encoded in UTF-8 unless explicitly defined.
This rule will be applied to all version of this protocol.

1.5 API versions

1.5.1 Supported API versions on firmware versions

Firmware Version	Supported API version	http	https
v1.3.0	v1	✓	-
v1.4.0	v1	✓	-
v1.4.0	v1	✓	-
v1.5.0	v1	✓	-
v1.6.0	v1	✓	-
v2.0.1	v2.0	✓	-
v2.1.0	v2.1	✓	-
v3.0.1	v2.2	✓	-
v3.1.0	v2.3	✓	-
V3.3.0	v2.3	✓	-
V3.4.0	v2.3	✓	✓

1.5.2 Difference between v2.0 and v2.1

	Description
1	Add new command (v2/firmware/model)
2	Add multicast_number field to the response of v2/info/status
3	Add runmode parameter and status_result to the response of the following commands. ● v2/system/forcestop ● v2/pattern/play ● v2/pattern/stop ● v2/phone/sip_paging ● v2/phone/sip_cancel ● v2/auxin/play ● v2/auxin/stop Add status to the response of the following commands ● v2/system/forcestop See chapter 6.

1.5.3 Difference between v2.1 and v2.2

	Description
1	Add microphone category commands.
2	Add set_speaker_output command and get_speaker_output command to controlio category.
3	Add new broadcasting states to chapter 5.
4	Add “destination” parameter to “sip, and “dtmf” parameter and SIP to the following commands and supports “target” can be set to a SIP target. ● v2/pattern/play (10.5) ● v2/auxin/play (10.9) ● (NEW command) v2/microphone/play (10.11)
5	Extend interval and delay of v2/pattern/play (10.5) up to 99 seconds.
6	Extend duration of pattern/play (10.5) up to 36000 seconds. But upper limit in Pattern Setting is 3600 seconds.
7	Add “rec” in target to the following commands. ● v2/pattern/list (10.5.1) ● v2/pattern/info (10.5.2) “rec” denotes Recorderd Audio is assigned to a pattern.

1.5.4 Difference between v2.2 and v2.3

	Description
1	Add error codes (See 2.5)
2	Add media category commands.
3	Add multicast category commands.

2 Command format

URL syntax

Port 80

```
http://<ipaddress>/api/v2/<category>/<command>?param1=XXX&param2=YYY&…
```

changed Port

```
http://<ipaddress>:<port>/api/v2/<category>/<command>?param1=XXX&param2=YYY&…
```

Item	Description	
<port>	Default port is 80. ● Configurable by IP Setting Tool (which reboots automatically) ● Configurable at Network screen by browser (needs reboot manually) ● Web browser port and Remote API port are same. ● Valid setting range is 80, 10000-40000. ● Must be different from RTSP and VMS port setting.	
v2	v1 or v2 Firmware which supports protocol v2 supports protocol v1 too to keep backward compatibility.	
Authentication	Digest authentication	
<ipaddress>	IP address of IP-A1 series device	
<category>	Category to group commands. for example, system, firmware or etc. defined at 9.1	
<command>	Command name. defined at 9.2	
Input data	Upload firmware	Send by POST method as request body. in application/octet-stream format. There is no query string.
	Upload config file.	
	Upload internal audio	Send by POST method as request body. in application/octet-stream format. Query string is used to specify how to handle internal audio

	Other	Send by GET method as query string. In the example of URL syntax, param1 and param2 are parameter names and XXX and YYY are their values respectively. Use valid parameter and value based on each command description.	
Query string	Separator	?	Separator to divide command part and query string part
		&	Separator to divide parameters
		=	Separator to divide parameter name and value
	Order of parameters	Order of parameters is arbitrary. don't need to care the parameter orders.	
	Optional parameter	Optional parameter is arbitrary. If omitted, default value will be used. **There are exceptions on default action for some command. They will be explained on each command description.**	
Command response	Except for downloading config file	application/json	
	Downloading config file (success)	application/octet-stream	
	Downloading config file (failure case)	application/json	

2.1 Command request/response example1

Explains pattern play command as a standard command example.
The command response is in application/json format.
The order of json labels are not ensured.

If 'result' element is true, it means success. If false, it means failure.
'response' element has a child element or children elements which contain additional information.

Command lists refer to only child element or children elements. (In this example, 'status')

In this example, parameter 'pattern_number' is mandatory, and other parameters (playcount, interval, volume) are optional.

See 10.5 for more detail.

Command request

```
GET /api/v2/pattern/play?pattern_number=2&playcount=2&interval=2&volume=-10 HTTP/1.1
Host: 192.168.14.1
```

Command response

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 62

{
    "result" : true,
    "response": {
        "status": "idle",
    }
}
```

2.2 Command request/response example2

Note: This type of command is same as v1 and v2. explained as v1 command.

Explains downloading config file as an example.
Normal commands return in application/json format as command response, but this command returns application/octet-stream format on success.

This type of command is this command only now.
The max value of Content-Length at the command response is around 80MB, but it may be increased in the future firmware.

It returns in application/json format on failure.

Command Request

```
GET /api/v1/config/download HTTP/1.1
Host: 192.168.14.1
```

Command response

```
HTTP/1.1 200 OK
Content-Type: application/octet-stream
Content-Length: xxx

<config data as binary data>
```

2.3 Command request/response example3

Note: This type of command is same as v1 and v2. explained as v1 command.

Explains uploading firmware as an example.
Normal commands pass input parameters by query string, but this command passes input data (firmware binary) in application/octet-stream format.

The command response is in application/json format and it returns firmware version.

Same rule applies to uploading config file and it sends data in application/octet-stream format.

The value of Content-Length in the command is same as input file size.
The value of Content-Length in uploading config file command is same as config file size.

Command Request

```
POST /api/v1/firmware/upload HTTP/1.1
Host: 192.168.14.1
Connection: close
Content-Length: 17473536
Content-Type: application/octet-stream

<binary data>
```

Command response

```
HTTP/1.1 200 OK
Connection: close
Content-Type: application/json
Content-Length: 86

{
    “result” : true,
    "response": {
        "version" : "2.0.0",
    }
}
```

2.4 Command request/response example4

Explains an example of command failure.
Invalid pattern number **22** is passed to pattern play command. It returns invalid parameter error.
See 2.5 for the list of error codes.

code element offers detail error information.
message element is only for users, not for software. software shouldn't use it on **message** element.

Command Request

```
GET /api/v2/pattern/play?pattern_number=22 HTTP/1.1
Host: 192.168.14.1
```

Command response

```
HTTP/1.1 400 Bad Request
Content-Type: application/json
Content-Length: 92

{
  "response": {
    "code": 1,
    "message": "invalid parameter"
  },
  "result": false
}
```

2.5 Error codes

× = Not Applicable

"code"	"message"	"url"	Description	HTTP Status code	Available API Version
not exist	not exist	×	Success	200 OK	v1 or later
1	invalid parameter		Invalid parameter	400 Bad Request	v1 or later
2	No parameter	×	Mandatory parameter is missing	400 Bad Request	v1 or later
3	Unauthorized	×	Username or password is wrong or Basic Authentication is used.	401 Unauthorized	v1 or later
4	Forbidden	×	Authorization failed. No privilege	403 Forbidden	v1 or later
5	Not Found	×	Not implemented	404 Not Found	v1 or later
6	method not allowed	×	Unsupported HTTP method	405 Method Not Allowed	v1 or later
7	Internal Error	×	Internal error	500 Internal Server Error	v1 or later
8	Not Implemented	×	Unsupported protocol version	501 Not Implemented	v1 or later
9	Service Unavailable	×	Busy, or feature is disabled by configuration	503 Service Unavailable	v2.0 or later
10	Request Timeout	×	Request timeout	504 Gateway Timeout	v2.0 or later
11	Disabled on recovery	×	Disabled in Recovery mode	403 Forbidden	v2.0 or later
12	Media file busy	×	Media file is busy	503 Service Unavailable	v2.0 or later
13	Not enough disk space available	×	Not enough space available	400 Bad Request	v2.3 or later
14	Media file size is too big	×	Media file size is too big	400 Bad Request	v2.3 or later
15	Only application/octet-stream is supported	×	Only application/octet-stream is supported Note: If invalid file is uploaded as audio data, invalid parameter will be returned.	415 Unsupported Media Type	v2.3 or later
16	Multicast Channel conflict	×	Multicast channel settings are conflicted	400 Bad Request	v2.3 or later
17	Redirect	<https://<ip>:<port>/request_uri	HTTP request redirected to HTTPS	307 Temporary Redirect	v2.4 or later

3 Compatibility

Firmware which is compatible with this specification supports Remote API version 1 for backward compatibility. Applications or scripts which use Remote API version 1 can work without change.

Remote API will be extended at feature enhancement like Ver 2.1, 2.2, or so.

API Version can be acquired by /api/v2/firmware/version.
API Version has nothing to do with firmware version.
When firmware is updated, API Version will not be incremented if there is no API change.

Difference between API Versions

Items	API Version 1	API Version 2
Unknown parameter	Ignore	Parameter error
More than two parameters in the same setting	Not supported	Parameter error

The following commands are removed on firmware which supports API v2.

- v1/system/settime
- v1/system/gettime
- v1/system/settimezone
- v1/system/gettimezone
- v1/controlio/getnumcin
- v1/controlio/getcin
- v1/controlio/getnumcin
- v1/controlio/getnumcout
- v1/controlio/getcout
- v1/controlio/setcout

Category **controlio** is officially supported on API v2.

4 Supported command on each model

	category	command	Supported API Version	Models				
				IP-A1SC15 IP-A1PC238 IP-A1PC580R/S	IP-A1AF	IP-A1PA12	IP-A1PG	IP-A1RM
1	System	reboot	v1 or later	○	○	○	○	○
		forcestop	v1 or later	○	○	○	○	○
2	firmware	upload	v1 or later	○	○	○	○	○
		update	v1 or later	○	○	○	○	○
		status	v1 or later	○	○	○	○	○
		version	v1 or later	○	○	○	○	○
		timeinfo	v1 or later	○	○	○	○	○
		model	v2.1 or later	○	○	○	○	○
3	info	command	v1 or later	○	○	○	○	○
		status	v1 or later	○	○	○	○	○
		loginpage	v1 or later	○	○	○	○	○
4	config	upload	v1 or later	○	○	○	○	○
		update	v1 or later	○	○	○	○	○
		download	v1 or later	○	○	○	○	○
		timeinfo	v1 or later	○	○	○	○	○
5	pattern	play	v1 or later	○	○	○	○	○
		stop	v1 or later	○	○	○	○	○
		timerstop	v2.0 or later	○	○	○	×	×
		list	v1 or later	○	○	○	○	○
		info	v1 or later	○	○	○	○	○

	category	command	Supported API version	Models				
				IP-A1SC15 IP-A1PC238 IP-A1PC580R/S	IP-A1AF	IP-A1PA12	IP-A1PG	IP-A1RM
6	volume	set_master	v1 or later	○	○	○	×	×
		get_master	v1 or later	○	○	○	×	×
		inc_master	v1 or later	○	○	○	×	×
		dec_master	v1 or later	○	○	○	×	×
		set_master_offset	v1 or later	○	○	○	×	×
		get_master_offset	v1 or later	○	○	○	×	×
		set_master_mute	v2.0 or later	○	○	○	×	×
		get_master_mute	v2.0 or later	○	○	○	×	×
7	phone	sip_paging	v1 or later	○	○	○	×	×
		sip_cancel	v1 or later	○	○	○	×	×
8	controlio	getnumcout	v2.0 or later	○	○	○	○	○
		setcout	v2.0 or later	○	○	○	○	○
		getcout	v2.0 or later	○	○	○	○	○
		getnumcin	v2.0 or later	○	○	○	○	○
		getcin	v2.0 or later	○	○	○	○	○
		set_speaker_output	v2.2 or later	×	×	○	×	×
		get_speaker_output	v2.2 or later	×	×	○	×	×
9	auxin	play	v2.0 or later	×	○	○	○	○
		stop	v2.0 or later	×	○	○	○	○
10	command	send	v2.0 or later	×	×	×	○	○
11	microphone	play	v2.2 or later	×	×	×	×	○
		stop	v2.2 or later	×	×	×	×	○
12	media	upload	v2.3 or later	○	○	○	○	○
13	multicast	set_receive_channels	v2.3 or later	○	○	○	×	×
		get_receive_channels	v2.3 or later	○	○	○	×	×

5 Operating status on each model

Explains supported operating status on each model.

Operating status can be acquired by /api/v1/info/status or /api/v2/info/status (See 10.3)

See 10.3.2.1 for auxiliary information on the status command.

Legend
API v2.2 or later
API v2 or later
API v1

	Operating status ("status" at info/status command)	Description	Supported API version	Models			
				IP-A1SC15 IP-A1PC238 IP-A1PC580R/S	IP-A1AF IP-A1PA12	IP-A1PG	IP-A1RM
1	idle	Standby state	v1 or later	○	○	○	○
2	sip_calling	Call to a SIP device in progress	v1 or later	○	○	×	×
3	sip_paging	SIP broadcasting in progress	v1 or later	○	○	×	×
4	multicast_paging	Multicast broadcasting in progress.	v1 or later	○	○	×	×
5	internal_audio_playing	Internal sound source broadcasting in progress. (Displays the pattern number and the sound source filename currently being executed.)	v1 or later	○	○	○	○
6	vms_broadcasting	VMS broadcasting in progress	v1 or later	○	○	×	×
7	local_broadcasting	Local broadcasting in progress	v2.0 or later	×	○	×	×
8	auxin_to_multicast	Input audio multicast broadcasting in progress	v2.0 or later	×	×	○	○
9	broadcast_disable	Transmitting device broadcasting suspended	v2.0 or later	×	×	○	○
10	system_mute	System currently being muted (by Mute control enabled by transmitter).	v2.0 or later	○	○	○	○
11	sip_dtmf_waiting	Waiting for SIP DTMF input	v2.0 or later	×	×	○	×
12	sip_to_multicast	SIP multicast broadcasting in progress	v2.0 or later	×	×	○	×
13	vms_to_multicast	VMS multicast broadcasting in progress	v2.0 or later	×	×	○	×
14	pattern_to_multicast	Multicast broadcasting of preset sound source pattern in progress	v2.0 or later	×	×	○	○

	Operating status ("status" at info/status command)	Description	Supported API version	Models			
				IP-A1SC15 IP-A1PC238 IP-A1PC580R/S	IP-A1AF IP-A1PA12	IP-A1PG	IP-A1RM
15	microphone_to_multicast	Microphone to Multicast	v2.2 or later	×	×	×	○
16	recorded_audio_to_multicast	Recorded Audio to Multicast	v2.2 or later	×	×	×	○
17	pattern_to_sip_calling	Pattern to SIP Calling	v2.2 or later	×	×	×	○
18	aux_to_sip_calling	AUX to SIP Calling	v2.2 or later	×	×	×	○
19	microphone_to_sip_calling	Microphone to SIP Calling	v2.2 or later	×	×	×	○
20	recorded_audio_to_sip_calling	Recorded Audio to SIP Calling	v2.2 or later	×	×	×	○
21	pattern_to_sip	Pattern to SIP	v2.2 or later	×	×	×	○
22	aux_to_sip	AUX to SIP	v2.2 or later	×	×	×	○
23	microphone_to_sip	Microphone to SIP	v2.2 or later	×	×	×	○
24	recorded_audio_to_sip	Recorded Audio to SIP	v2.2 or later	×	×	×	○

6 Sync response option (API v2.1 or later)

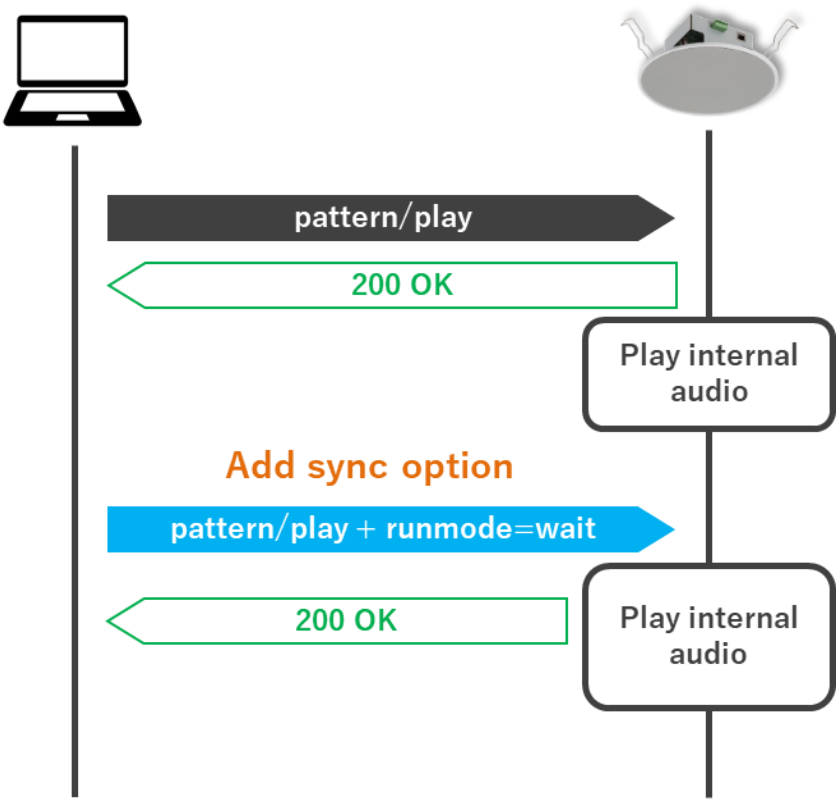
Run command which start/stop broadcasting synchronously by using sync response option (runmode=wait).

If higher priority or same priority broadcast is running, the command will fail with 503 error code (See 2.5)

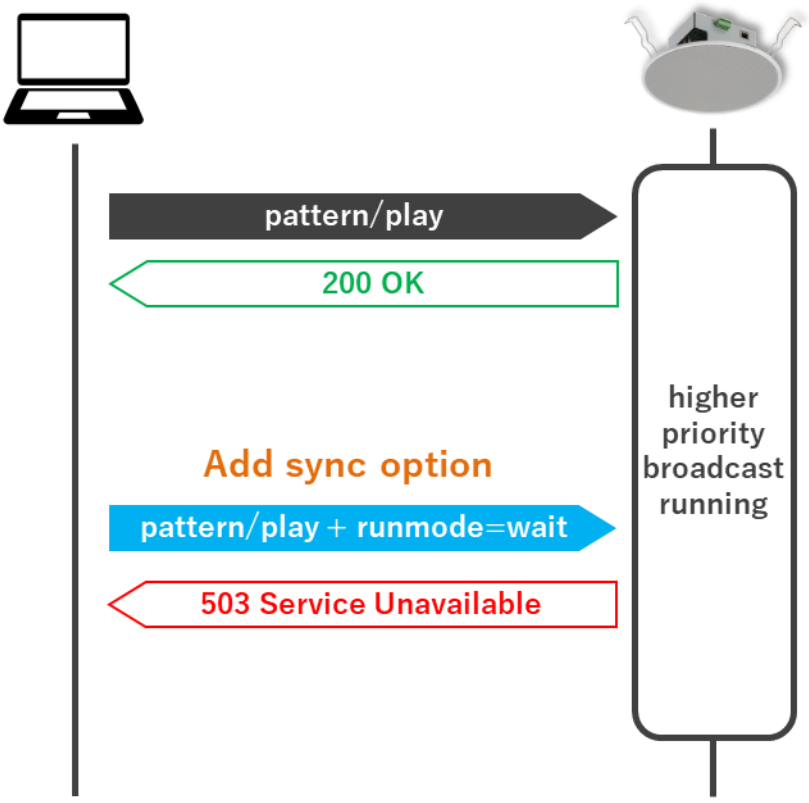
Related commands

category	command	API vesion
system	forcestop	v2.1 or later
pattern	play	v2.1 or later
	stop	v2.1 or later
phone	sip_paging	v2.1 or later
	sip_cancel	v2.1 or later
auxin	play	v2.1 or later
	stop	v2.1 or later
microphone	play	v2.2 or later
	stop	v2.2 or later
media	upload	v2.3 or later

- Response to play



- Response not to play because of higher priority broadcast



7 DTMF option (API v2.2 or later)

related commands

category	command
pattern	play
auxin	play
microphone	play

Characters	How to specify
0 ～ 9	specify as it is
a, b, c, d	specify as it is
A, B, C, D	specify as it is
*	specify as it is
#	H or %23

Note: H means hash tag. #23 is URL encoding of #.

Example

Source	Destination
192.168.14.1	192.168.14.2

DTMF	URL
AB	http://192.168.14.1/api/v2/microphone/play?destination=sip&target=192.168.14.2&dtmf=AB
#1	http://192.168.14.1/api/v2/microphone/play?destination=sip&target=192.168.14.2&dtmf=%231
#*	http://192.168.14.1/api/v2/microphone/play?destination=sip&target=192.168.14.2&dtmf=H*

8 Number of input ports / output ports on each model

Port numbers on each model

		IP-A1SC15 IP-A1PC238 IP-A1PC580R IP-A1PC580S IP-A1AF IP-A1RM	IP-A1PG
1	number of input ports	2	4
2	number of output ports	1	1

Note: Controlling input/output ports are supported on v2.0 or later.

9 Supported commands

Each model supports different command by model.
See chapter 4 in each model.

All supported commands can be acquired by **/api/v2/info/command**.
It returns category and command information.

When new categories or new commands are added, supported categories and commands can be confirmed by the command.

9.1 Category

Category	Description	Available API Version
system	Control device itself	v1 or later
firmware	Update firmware	v1 or later
info	Get information	v1 or later
config	Upload/download config file	v1 or later
pattern	Start/stop pattern, get information	v1 or later
volume	Set/get volume	v1 or later
phone	Start/stop SIP	v1 or later
controlio	Set/get control I/O ports	v2.0 or later
auxin	Start/stop LINE/MIC input multicast broadcast, local broadcasting or AUX IN to individual SIP broadcasting	v2.0 or later
command	Send HTTP Command Sets	v2.0 or later
microphone	Start/Stop Broadcast from Microphone	v2.2 or later
media	Upload a media file	v2.3 or later
multicast	Set/get multicast channel	v2.3 or later

9.2 Command

9.2.1 Category “system“

Category	Command	Description	Available API Version
system	reboot	Reboot a device	v1 or later
system	forcestop	Abort broadcasting	v1 or later

9.2.2 Category “firmware“

Category	Command	Description	Available API Version
firmware	upload	Upload firmware	v1 or later
firmware	update	Start updating firmware	v1 or later
firmware	status	Get the status of updating firmware	v1 or later
firmware	version	Get the version of firmware	v1 or later (extended at v2.0)
firmware	timeinfo	Get timeout information on firmware update	v1 or later
firmware	model	Get model information	v2.1 or later

9.2.3 Category “info“

Category	Command	Description	Available API Version
info	command	Get information of supported commands.	v1 or later (extended at v2.0)
info	status	Get operating status information	v1 or later (extended at v2.0)
info	loginpage	Get login URL	v1 or later

9.2.4 Category “config“

Category	Command	Description	Available API Version
config	download	Download config file	v1 or later
config	upload	Upload config file	v1 or later
config	update	Update config file	v1 or later
config	timeinfo	Get timeout information on config file update	v1 or later

9.2.5 Category “pattern“

Category	Command	Description	Available API Version
pattern	play	Play a pattern	v1 or later (extended at v2.0)
pattern	stop	Stop a pattern	v1 or later (extended at v2.0)
pattern	timerstop	Cancel pattern of timer type	v2.0 or later
pattern	list	Get information of all patterns	v1 or later (extended at v2.0)
pattern	info	Get information of a pattern	v1 or later (extended at v2.0)

9.2.6 Category “volume“

Category	Command	Description	Available API Version
volume	set_master	Set master volume	v1 or later
volume	get_master	Get master volume and range	v1 or later
volume	inc_master	Increment master volume	v1 or later
volume	dec_master	Decrement master volume	v1 or later
volume	set_master_offset	Set master volume offset	v1 or later
volume	get_master_offset	Get master volume offset and range	v1 or later
volume	set_master_mute	Set mute and unmute	v2.0 or later
volume	get_master_mute	Get mute state	v2.0 or later

9.2.7 Category “phone“

Category	Command	Description	Available API Version
phone	sip_paging	Start SIP broadcasting	v1 or later
phone	sip_cancel	Stop SIP broadcasting	v1 or later

9.2.8 Category “controlio“

Category	Command	Description	Available API Version
controlio	getnumcout	Get number of output pins	v2.0 or later
controlio	setcout	Get output pin state	v2.0 or later
controlio	getcout	Get output pin state	v2.0 or later
controlio	getnumcin	Get number of input pins	v2.0 or later
controlio	getcin	Get input pin state	v2.0 or later
controlio	set_speaker_output	Switch speaker source (IP-A1PA12 only)	v2.2 or later
controlio	get_speaker_output	Get internal/external speaker source. (IP-A1PA12 only) Not only set_speaker_output but also 24V DC signal cut can change speaker source to external. This command can get the state in both cases.	v2.2 or later

9.2.9 Category “auxin“

Category	Command	Description	Available API Version
auxin	play	Start LINE/MIC input multicast broadcast, local broadcasting or AUX IN to individual SIP broadcasting	v2.0 or later
auxin	stop	Stop LINE/MIC input multicast broadcast, local broadcasting or AUX IN to individual SIP broadcasting	v2.0 or later

9.2.10 Category “command“

Category	Command	Description	Available API Version
command	send	Send HTTP Command Sets	v2.0 or later

9.2.11 Category “microphone“

Category	Command	Description	Available API Version
microphone	play	Start Broadcast from microphone.	v2.2 or later
microphone	stop	Stop Broadcast from microphone	v2.2 or later

9.2.12 Category “media“

Category	Command	Description	Available API Version
media	upload	Upload a media file	v2.3 or later

9.2.13 Category “multicast“

Category	Command	Description	Available API Version
multicast	set_receive_channels	Set receiving multicast channels	v2.3 or later
multicast	get_receive_channels	Get receiving multicast channels	v2.3 or later

10 Input / Output parameters of supported commands

10.1 Category “system“

See 9.2.1 for command meaning

URL Syntax

http://<ipaddress>/api/v1/system/<command>

URL Syntax

http://<ipaddress>/api/v2/system/<command>

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
reboot	none	-	v1 or later
forcestop (command itself v1 or later)	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.1 or later

Output parameters

<command>	Command response	Available API Version
reboot	none	v1 or later
forcestop	status broadcasting state before issuing the command. (see 5 and 10.3.2)	v2.1 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait.	v2.1 or later

10.2 Category “firmware“

See 9.2.2 for command meaning

URL Syntax

http://<ipaddress>/api/v1/firmware/<command>

URL Syntax

http://<ipaddress>/api/v2/firmware/<command>

Input parameters

<command>	parameter	Available API Version
upload	Upload firmware date in application/octet-stream format by POST method	v1 or later
update	None After issuing the command, the firmware will be updated and rebooted.	v1 or later
status	None	v1 or later
version	None	v1 or later
timeinfo	None	v1 or later
model	None	v2.1 or later

Output parameters

<command>	Command response	Available API Version
upload	version Uploaded firmware version	v1 or later
update	None	v1 or later
status	status Status of firmware update “success” : on success “progress” : on updating “fail” : on failure	v1 or later
version	version String separated by period like “2.1.0”	v1 or later
	api_version String separated by period like “2.1”	v2 or later
timeinfo	fwupload_timeout Timeout time of firmware upload command in seconds	v1 or later
	fwupdate_timeout Timeout time of firmware update command in seconds	v1 or later
	fwupdate_first_check Delay to check firmware update status from completing firmware update command in seconds	v1 or later
	fwupdate_done_timeout Timeout from completing firmware update command in seconds	v1 or later

	fwupdate_first_check Delay to check firmware update status from completing firmware update command in seconds	v1 or later
model	version String separated by period like “2.1.0”	v2.1 or later
	api_version String separated by period like “2.1”	v2.1 or later
	model_name Model name (Product name) ex: IP-A1PG"	v2.1 or later
	model_type type of model "transmitter" or "receiver"	v2.1 or later
	terminal_name Terminal name which is set on the Network settings screen by browser or on Network Setting by IP Setting Tool. ex: "My-IP-A1PG"	v2.1 or later
	mac_address MAC Address ex:"00-05-F9-FF-A7-D1"	v2.1 or later
	hw_version H/W Revision as string ex:"1"	v2.1 or later

10.3 Category “info“

See 9.2.3 for command meaning

URL SYNTAX

http://<ipaddress>/api/v1/info/<command>

URL SYNTAX

http://<ipaddress>/api/v2/info/<command>

Input parameters

<command>	Parameter	Available API Version
command	None	v1 or later
status	None	v1 or later
loginpage	None	v1 or later

Output parameters

<command>	Command response	Available API Version
command	See 10.3.1 for detail The command output depends on models because each model implements different feature.	v1 or later (extended at v2.0)
status	status operating status string (See 5 and 10.3.2)	v1 or later
	pattern_number Pattern number in play. Valid range: 1 – 20. 0 when pattern is not playing.	v1 or later

	destination_number Multicast destination number. Valid range: 1 – 20. Only exists on a model which supports multicast sending. 0 when multicast is not sending.	v2.0 or later
	multicast_number Multicast channel number Only exists on a receiver model Valid range: 1 – 20. 0 when multicast is not broadcasting.	v2.1 or later
loginpage	loginpage Login URL	v1 or later

10.3.1 Command “command”

The response of “**command**” command

Structure

- Category name keys exist under response.
- Command keys exist under a category key.

```
HTTP/1.1 200 OK
Server: xxxxxxxxxx
Content-Type: application/json
Content-Length: xxx

{
  "response": {
    "auxin": {
      "play": {
        …
      },
      "stop": {
        …
      }
    },
    "config": {
      "download": {
        …
      },
       …
    },
    …
  },
  "result": true
}
```

Command definition structure

Top level	Second level	Third level	Value
method			Supported HTTP methods array such as “GET” or “POST”
params Note: If a command has no parameters, this JSON object will be empty.	Supported parameters	**mandatory**	true or false true if mandatory parameter false if optional parameter
		type	One of the following values “values” “range” "custom" “values” means the parameter can be specified from pre-defined list. “range” means the parameter can be set from the range. "custom" means the parameter is custom parameter.
		values	Returns all possible values.
		range	Returns min and max in array. The first item is min The second item is max.
privilege			“user” or “admin” Necessary user privilege to run the command.

This is an example of auxin/play command (on IP-A1AF)

```
"auxin": {
  "play": {
    "method": ["GET"],
    "params": {
      "destination": {
        "mandatory": false,
        "type": "values",
        "values": ["speaker"]
      }
    },
    "privilege": "user"
  },
```

Meaning

- Available HTTP method GET
- Valid parameter destination
- destination parameter (on IP-A1AF) optional parameter
  (on IP-A1AF) valid value is “speaker”
- Privilege user

This is an example of pattern/info command

```
"pattern": {
  "info": {
    "method": [
      "GET"
    ],
    "params": {
      "pattern_number": {
        "mandatory": true,
        "range": [
          1,
          20
        ],
        "type": "range"
      }
    },
    "privilege": "user"
  },
```

Meaning

- Available HTTP method GET
- Valid parameter pattern_number
- pattern_number parameter mandatory parameter
  valid range is from 1 to 20.
- Privilege user

This is an example of pattern/list command

```
"pattern": {
  "list": {
    "method": [
      "GET"
    ],
    "params": {},
    "privilege": "user"
  },
```

Meaning

- Available HTTP method GET
- Valid parameter none
- Privilege user

10.3.2 status “command”

Output parameters

Command response	Available API Version
status operating status string (See chapter 5)	v1 or later
pattern_number Pattern number in play. Valid range: 1 – 20. 0 when not playing.	v1 or later
destination_number Multicast destination number. Valid range: 1 – 20. Only exists on a model which supports multicast sending. 0 when not sending multicast.	v2.0 or later
multicast_number Multicast channel number Only exists on a receiver model Valid range: 1 – 20. 0 when not multicast playing.	v2.1 or later

The response of “**status**” command

See 10.3.2.1 for the difference between models.

pattern_number means pattern number which is currently playing.
If pattern is not playing, it is 0.

destination_number field only exists on IP-A1PG/IP-A1RM only.
It means multicast group channel.
If multicast is not sending, it is 0.

multicast _number field only exists on receiver models
multicast_number means multicast channel which is currently playing.
If multicast is not playing, it is 0.

Example: Idle on IP-A1AF

This is an example response for idle state on IP-A1AF.

```
HTTP/1.1 200 OK
Server: xxxxxxxxxx
Content-Type: application/json
Content-Length: xxx

{
   "response": {
      “multicast_number”: 0,
      "pattern_number": 0,
      "status": "idle"
   },
   "result": true
}
```

multicast_number field exists on API v2.1 or later.

Example: Idle on IP-A1PG

This is an example response for idle state on IP-A1PG.

```
HTTP/1.1 200 OK
Server: xxxxxxxxxx
Content-Type: application/json
Content-Length: xxx

{
  "response": {
    "destination_number": 0,
    "pattern_number": 0,
    "status": "idle"
  },
  "result": true
}
```

Example: internal audio playing on IP-A1AF

This is an example response for playing internal audio on IP-A1AF.
In this example, pattern 1 is playing.

```
HTTP/1.1 200 OK
Server: xxxxxxxxxx
Content-Type: application/json
Content-Length: xxx

{
  "response": {
    “multicast_number”: 0,
    "pattern_number": 1,
    "status": "internal_audio_playing"
  },
  "result": true
}
```

multicast_number field exists on API v2.1 or later.

Example: multicast broadcasting on IP-A1AF

This is an example response for broadcasting multicast on IP-A1AF.
In this example, multicast 1 is broadcasting.

```
HTTP/1.1 200 OK
Server: xxxxxxxxxx
Content-Type: application/json
Content-Length: xxx

{
  "response": {
    “multicast_number”: 1,
    "pattern_number": 0,
    "status": "multicast_paging"
  },
  "result": true
}
```

multicast_number field exists on API v2.1 or later.

Example: Multicast broadcasting on IP-A1PG

This is an example response for Multicast broadcasting on IP-A1PG.
In this example, pattern 1 is sending to multicast group 2.

```
HTTP/1.1 200 OK
Server: xxxxxxxxxx
Content-Type: application/json
Content-Length: xxx

{
  "response": {
    "destination_number": 2,
    "pattern_number": 1,
    "status": "pattern_to_multicast"
  },
  "result": true
}
```

10.3.2.1 auxiliary information in the status command

Legend
Changed at v2 FW
Same as v1 FW

Legend	meaning
n	Pattern Number
m	Multicast Channel
-	Not supported

Auxiliary information on broadcast information.

Explain pattern_number and destination_number, multicast_number in responses in status command in the table at the next page.

	Operating status ("status" at info/status command)	State at status screen	pattern_number (v1 or later)				destination_number (v2 or later)				multicast_number (v2.1 or later)			
			IP-A1SC15 IP-A1PC238 IP-A1PC580R/S	IP-A1AF IP-A1PA12	IP-A1PG	IP-A1RM	IP-A1SC15 IP-A1PC238 IP-A1PC580R/S	IP-A1AF IP-A1PA12	IP-A1PG	IP-A1RM	IP-A1SC15 IP-A1PC238	IP-A1AF IP-A1PA12	IP-A1PG	IP-A1RM
1	idle	Idle	0	0	0	0	-	-	0	0	-	-	-	-
2	sip_calling	SIP Calling	0	0	0	-	-	-	-	-	-	-	-	-
3	sip_paging	SIP	0	0	0	-	-	-	-	-	-	-	-	-
4	multicast_paging	Multicast m	0	0	0	-	-	-	-	-	m	m	-	-
5	internal_audio_playing	Pattern n (XXX)	n	n	n	n	-	-	-	-	-	-	-	-
6	vms_broadcasting	VMS Broadcasting	0	0	0	0	-	-	-	-	-	-	-	-
7	local_broadcasting	Local Broadcasting	-	0	0	-	-	-	-	-	-	-	-	-
8	auxin_to_multicast	LINE/MIC IN to to Multicast (YYY)	-	-	-	0	-	-	m	m	-	-	-	-
9	broadcast_disable	Broadcast Disable	-	-	-	0	-	-	0	0	-	-	-	-
10	system_mute	System Mute	0	0	0	0	-	-	0	0	-	-	-	-
11	sip_dtmf_waiting	SIP DTMF Waiting	-	-	-	0	-	-	0	0	-	-	-	-
12	sip_to_multicast	SIP to Multicast	-	-	-	0	-	-	m	m	-	-	-	-
13	vms_to_multicast	VMS to Multicast	-	-	-	0	-	-	m	m	-	-	-	-
14	pattern_to_multicast	Pattern n (XXX) to Multicast (YYY)	-	-	-	n	-	-	m	m	-	-	-	-

	Operating status ("status" at info/status command)	State at status screen	pattern_number (v2.2 or later)				destination_number (v2 or later)				multicast_number (v2.1 or later)			
			IP-A1SC15 IP-A1PC238 IP-A1PC580R/S	IP-A1AF IP-A1PA12	IP-A1PG	IP-A1RM	IP-A1SC15 IP-A1PC238 IP-A1PC580R/S	IP-A1AF IP-A1PA12	IP-A1PG	IP-A1RM	IP-A1SC15 IP-A1PC238 IP-A1PC580R/S	IP-A1AF IP-A1PA12	IP-A1PG	IP-A1RM
15	microphone_to_multicast	Microphone to Multicast	-	-	-	-	-	-	-	-	-	-	-	-
16	recorded_audio_to_multicast	Recorded Audio to Multicast	-	-	-	-	-	-	-	-	-	-	-	-
17	pattern_to_sip_calling	Pattern to SIP Calling	-	-	-	n	-	-	-	m	-	-	-	-
18	aux_to_sip_calling	AUX to SIP Calling	-	-	-	-	-	-	-	-	-	-	-	-
19	microphone_to_sip_calling	Microphone to SIP Calling	-	-	-	-	-	-	-	-	-	-	-	-
20	recorded_audio_to_sip_calling	Recorded Audio to SIP Calling	-	-	-	-	-	-	-	-	-	-	-	-
21	pattern_to_sip	Pattern to SIP	-	-	-	n	-	-	-	m	-	-	-	-
22	aux_to_sip	AUX to SIP	-	-	-	-	-	-	-	-	-	-	-	-
23	microphone_to_sip	Microphone to SIP	-	-	-	-	-	-	-	-	-	-	-	-
24	recorded_audio_to_sip	Recorded Audio to SIP	-	-	-	-	-	-	-	-	-	-	-	-

10.4 Category “config“

See9.2.4 for command meaning

URL Syntax

```
http://<ipaddress>/api/v1/config/<command>
```

URL Syntax

```
http://<ipaddress>/api/v2/config/<command>
```

Input parameters

<command>	Parameter	Available API Version
download	none	v1 or later
upload	Send config data in application/octet-stream format by POST method	v1 or later
update	none After issuing the command, the config file will be updated and rebooted.	v1 or later
timeinfo	none	v1 or later

Output parameters

<command>	Command response	Available API Version
download	Returns config file data in application/octet-stream format	v1 or later
upload	none	v1 or later
update	none	v1 or later
timeinfo	configupload_timeout timeout for uploading config file (seconds)	v1 or later
	configupdate_timeout timeout for updating config file (seconds)	v1 or later
	configdownload_timeout timeout for downloading config file (seconds)	v1 or later

10.5 Category “pattern“

See 9.2.5 for command meaning

URL Syntax

http://<ipaddress>/api/v1/pattern/<command>?param1=XXX& param2=YYY&…

URL Syntax

http://<ipaddress>/api/v2/pattern/<command>?param1=XXX& param2=YYY&…

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
play	pattern_number pattern number to be played. Setting range: 1 to 20 pattern to be set to timer type can be controlled by this command. If run, returns 503 Service Unavailable (code 9) at 2.5	M	v1 or later
	volume volume offset to be played. Setting range: −20 to +20 dB If omitted: 0 is assumed. This value is appended to the setting at PATTERN REGISTRATION SCREEN. The setting will be valid once.	O	v1 or later

	playcount playing count Setting range: 1 – 10 (times) If omitted: PATTERN REGISTRATION SCREEN setting will be used. **mutually exclusive with duration** override the setting at pattern screen. The setting will be valid once.	O	v1 or later
	interval playing interval. Setting range: 0 – 99 (seconds) If omitted: PATTERN REGISTRATION SCREEN setting will be used. override the setting at pattern screen. The setting will be valid once.	O	v1 or later(up to 60) v2.2. or later(up to 99)
	duration playing time Setting range: 5 – 36000 (seconds) If omitted: PATTERN REGISTRATION SCREEN setting will be used. **mutually exclusive with playcount** override the setting at pattern screen. The setting will be valid once. **If destination param is “sip”, Call Time Limit on SIP screen is upper limit of this parameter.**	O	v2.0 or later (up to 3600) v2.2 or later (up to 36000)

	destination destination of broadcasting or sending Valid values ● speaker (on a receiver) ● multicast1 (on a sender) … ● multicast20 (on a sender) ● sip (on Remote Mic)	Receiver: O Sender: M	v2.0 or later v2.2 or later(“sip”)
	target call target IP Address or SIP User ID when destination is “sip”. Ex: 192.168.14.2	Remote Mic: O	v2.2 or later
	dtmf dtmf to send from Remote Mic Ex: “01” See chapter 7.	Remote Mic: O	v2.2 or later
	delay Delay time to be played. Setting range: 0 – 99 (seconds) If omitted: PATTERN REGISTRATION SCREEN setting will be used. Start output control out, and start playing after this setting. override the setting at pattern screen. The setting will be valid once.	O	v2.0 or later(up to 30) v2.2 or later(up to 99)

	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.1 or later
stop	pattern_number pattern number to be stopped. Setting range: 1 – 20 pattern to be set to timer type can be controlled by this command. If run, returns 503 Service Unavailable (code 9) at 2.5 (v2 or later) If omitted, detect running pattern automatically. **For v1, pattern_number is mandatory, can't be omitted.**	v1: M v2: O	v1 or later
	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.1 or later

timerstop	pattern_number pattern number to be stopped. Setting range: 1 – 20 pattern to be set to non-timer type can be controlled by this command. If run, returns 503 Service Unavailable (code 9) at 2.5	M	v2.0 or later
list	none	-	v1 or later
info	pattern_number pattern number to be got. Setting range: 1 – 20	M	v1 or later

Output parameters

<command>	Command response		Available API Version
play	status broadcasting state before issuing the command. (see 5 and 10.3.2)		v1 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait..		v2.1 or later
stop	status broadcasting state before issuing the command. (see 5 and 10.3.2)		v1 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait..		v2.1 or later
timerstop	status broadcasting state before issuing the command. (see 5 and 10.3.2)		v2.0 or later
list	patternYYY YYY= 1 to 20 (all)	operation_type “media_playback”	v1 or later
		name pattern name	v1 or later
		target internal audio file name or “rec” for Recorded Audio.	v1 or later

		playcount play count Setting range: 1 – 10 (times)	v1 or later
		interval interval time Setting range: 0 – 99 (seconds)	v1 or later (up to 60) v2.2 or later (up to 99)
		volume volume to be played. Setting range: −20 to +20 dB	v1 or later
		delay delay time Setting range: 0 – 99 (seconds)	v2.0 or later (up to 60) v2.2. or later (up to 99)
		repeat Repeat setting at pattern screen. One of the following values. playcount (Play Count) duration (Duration) timer (Timer)	v2.0 or later
info	patternYYY YYY= 1 to 20 (one of them)	operation_type “media_playback”	v1 or later
		name pattern name	v1 or later
		target internal audio file name or “rec” for Recorded Audio.	v1 or later
		playcount play count Setting range: 1 – 10 (times)	v1 or later

		interval interval time Setting range: 0 – 99(seconds)	v1 or later (up to 60) v2.2 or later (up to 99)
		volume volume to be played Setting range: −20 to +20 dB	v1 or later
		delay delay time Setting range: 0 – 99 (seconds)	v2.0 or later (up to 30) v2.2. or later (up to 99)
		repeat Repeat setting at pattern screen. One of the following values. playcount (Play Count) duration (Duration) timer (Timer)	v2.0 or later

10.5.1 Command “list“

An example of list command response

```
HTTP/1.1 200 OK
Server: xxxxxxxxxx
Content-Type: application/json
Content-Length: xxx

{
    "result" : true,
     "response": {
          "pattern1": {
               "name": "Pattern 1",
               "operation_type": "mediaplayback",
               "target" : "a.wav",
               "interval" : 10,
               "volume" : 0,
               "playcount" : 10,
               "duration": 60,
               "delay": 0,
               "repeat": "playcount"
          },
          "pattern2": {
               "name": "Pattern 2",
               "operation_type": "mediaplayback",
               "target" : "b.mp3",
               "interval" : 10,
               "volume" : 20,
               "playcount" : 3,
               "duration": 3600,
               "delay": 0,
               "repeat": "duration"
          },
          …
```

10.5.2 Command “info“

An example of info command response

```
HTTP/1.1 200 OK
Server: xxxxxxxxxx
Content-Type: application/json
Content-Length: xxx

{
    "result" : true,
     "response": {
          "pattern3": {
               "name": "Pattern 3",
               "operation_type": "mediaplayback",
               "target" : "b.mp3",
               "interval" : 1,
               "volume" : -20,
               "playcount" : 4,
               "duration": 120,
               "delay": 0,
               "repeat": "timer",
          }
     }
}
```

10.6 Category “volume“

See 9.2.6 for command meaning

URL Syntax

http://<ipaddress>/api/v1/volume/<command>?param1=XXX

URL Syntax

http://<ipaddress>/api/v2/volume/<command>?param1=XXX

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
set_master	volume volume to be set Setting Range: 0 (Mute) – 61 (0 dB) Initial setting: 31 (−30 dB) Connected with the setting at AUDIO SETTINGS SCREEN. Reflected immediately. Persistent after reboot.	M	v1 or later
get_master	none Connected with the display at AUDIO SETTINGS SCREEN.	-	v1 or later

inc_master	none Connected with the setting at AUDIO SETTINGS SCREEN. Reflected immediately. Persistent after reboot.	-	v1 or later
dec_master	none Connected with the setting at AUDIO SETTINGS SCREEN. Reflected immediately. Persistent after reboot.	-	v1 or later
set_master_offset	offset Offset volume correction to the Master Volume Setting range: −20 to +20 dB Initial setting: 0 (0 dB) Connected with the display at AUDIO SETTINGS SCREEN. Reflected immediately. Persistent after reboot. Can't change the setting at the screen. Remote API only can change this.	M	v1 or later
get_master_offset	none Connected with the display at AUDIO SETTINGS SCREEN.	-	v1 or later

set_master_mute	mute_state mute state to be set one of the following values. mute unmute Initial setting: unmute Connected with the setting at AUDIO SETTINGS SCREEN. Reflected immediately. Persistent after reboot.	M	v2.0 or later
get_master_mute	none Connected with the display at AUDIO SETTINGS SCREEN.	-	v2.0 or later

Output parameters

<command>	Command response	Available API Version
set_master	volume same as input parameter	v1 or later
get_master	volume current Master Volume	v1 or later
	max max Master Volume	v1 or later
	min min Master Volume	v1 or later
inc_master	volume new master volume	v1 or later
dec_master	volume new master volume	v1 or later
set_master_offset	offset same as input parameter	v1 or later
get_master_offset	offset current offset volume correction to the Master Volume	v1 or later
	min min of offset volume correction to the Master Volume	v1 or later
	max max of offset volume correction to the Master Volume	v1 or later

set_master_mute	mute_state new mute state One of the following values mute unmute	v2.0 or later
get_master_mute	mute_state current mute state One of the following values mute unmute	v2.0 or later

10.7 Category “phone“

See 9.2.6 for command meaning

URL Syntax

```
http://<ipaddress>/api/v1/phone/<command>?param1=XXX
```

URL Syntax

```
http://<ipaddress>/api/v2/phone/<command>?param1=XXX
```

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
sip_paging	target target IP address or target SIP user ID	M	v1 or later
	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.1 or later
sip_cancel	(command itself)	-	v1 or later

	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.1 or later

Output parameters

<command>	Command response	Available API Version
sip_paging	status broadcasting state before issuing the command. (see 5 and 10.3.2)	v1 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait..	v2.1 or later
sip_cancel	status broadcasting state before issuing the command. (see 5 and 10.3.2)	v1 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait..	v2.1 or later

10.8 Category “controlio“

See 9.2.8 for command meaning

URL Syntax

```
http://<ipaddress>/api/v2/controlio/<command>?param1=XXX
```

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
getnumcout	none	-	v2 or later
setcout	port_number output port number (from 1 to the number of ports) See chapter 8 for available ports on each model.	M	v2 or later
	value value to be set One of the following values on off	M	v2.0 or later
	delay delay time to output control-out (0 to 60) (seconds) If input parameter is omitted, 0 is assumed.	O	v2.0 or later
	duration duration time to keep on (0 to 360) valid only when value is on If input parameter is omitted, keep the state until changed.	O	v2.0 or later

getcout	port_number output port number to get. (from 1 to the number of ports) See chapter 8 for available ports on each model. If input parameter is omitted, returns all port information.	O	v2.0 or later
getnumcin	none	-	v2.0 or later
getcin	port_number input port number to be got. (from 1 to the number of ports) See chapter 8 for available ports on each model. If input parameter is omitted, returns all port information.	O	v2.0 or later
set_speaker_output (IP-A1PA12 Only)	speaker_output switch speaker output source. “internal” Internal Output “external” External Input.	M	v2.2 or later
get_speaker_output (IP-A1PA12 Only)	none	-	v2.2 or later

Output parameters

<command>	Command response	Available API Version
getnumcout	num_port number of output ports.	v2.0 or later
setcout	COUTxxx (xxx is an integer which means port number) One of the following values on off	v2.0 or later
getcout	COUTxxx (xxx is an integer which means port number) One of the following values on off If input parameter is omitted, returns all port information.	v2.0 or later
getnumcin	num_port number of output ports which depends on model. See chapter 8	v2.0 or later

getcin	CINxxx (xxx is an integer which means port number) One of the following values on off If input parameter **port_number** is omitted, returns all port information. All port information exists in CIN1, CIN2, … format.	v2.0 or later
set_speaker_output (IP-A1PA12 Only)	speaker_output same as input parameter as speaker_output	v2.2 or later
get_speaker_output (IP-A1PA12 Only)	speaker_output current speaker state. “internal” Internal Output “external” External Input.	v2.2 or later

10.9 Category “auxin“

See 9.2.9 for command meaning

URL Syntax

```
http://<ipaddress>/api/v2/auxin/<command>?param1=XXX
```

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
play	destination broadcast destination (local speaker or multicast destination) Valid values ● speaker (on IP-A1AF) ● multicast1 (on IP-A1PG/ IP-A1RM) … ● multicast20 (on IP-A1PG/ IP-A1RM) ● sip (on IP-A1RM) IP-A1AF ● If omitted, speaker is implied ● When setting is **speaker,** both speaker and LINE/OUT is active. ● Local Broadcasting Function must be enabled to use this API. IP-A1PG ● mandatory parameter IP-A1RM ● mandatory parameter	O(IP-A1AF) M(IP-A1PG) M(IP-A1RM)	v2.0 or later

	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.1 or later
	target call target IP Address or SIP User ID when destination is “sip”. Ex: 192.168.14.2	O	v2.2 or later
	dtmf dtmf to send from Remote Mic Ex: “01” See chapter 7.	O	v2.2 or later
stop	none IP-A1AF ● Local Broadcasting Function must be enabled to use this API.	-	v2.0 or later
	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.1 or later

Output parameters

<command>	Command response	Available API Version
play	status broadcasting state before issuing the command. (see 5 and 10.3.2)	v2.0 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait..	v2.1 or later
stop	status broadcasting state before issuing the command. (see 5 and 10.3.2)	v2.0 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait..	v2.1 or later

10.10 Category “command“

See 9.2.10 for command meaning

URL Syntax

```
http://<ipaddress>/api/v2/command/<command>?param1=XXX
```

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
send	number command set number to send (0 to 9)	M	v2.0 or later

Output parameters

<command>	Command response	Available API Version
send	number command set number which is same as input parameter (0 to 9)	v2.0 or later

10.11 Category “microphone“

See 9.2.11 for command meaning.

URL Syntax

```
http://<ipaddress>/api/v2/microphone/<command>?param1=XXX
```

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
play (IP-A1RM only)	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.2 or later
	destination broadcast destination (sip or multicast destination) Valid values ● multicast1 … ● multicast20 ● sip	M	v2.2 or later
	target target IP address or target SIP user ID	M	v2.2 or later

	dtmf dtmf to be sent. See chapter 7.	O	v2.2 or later
stop (IP-A1RM only)	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.2 or later

Output parameters

<command>	Command response	Available API Version
play	status broadcasting state before issuing the command. (see 5 and 10.3.2)	v2.2 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait.	v2.2 or later
stop	status broadcasting state before issuing the command. (see 5 and 10.3.2)	v2.2 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait..	v2.2 or later

10.12 Category “media“

See 9.2.12 for command meaning

URL Syntax

```
http://<ipaddress>/api/v2/firmware/<command>
```

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
upload	Upload audio data as request body in application/octet-stream format by POST method	M	v2.3 or later
	media_number number of media list to be uploaded Setting range: 1 to 20	M	v2.3 or later
	filename file name to be updated. (URL encoded string of UTF8) Setting range: same as media screen If omitted: automatically assign a filename	O	v2.3 or later
	check_busy specify the action when the media to be uploaded is in use. Setting range: 0 or 1 0: stop if the target media is playing, and update 1: Reject if the target media is playing	O	v2.3 or later

	pattern_number pattern number to be played after upload Setting range: 1 to 20 If the pattern is Timer, uploading will be rejected.	O	v2.3 or later
	runmode synchronous mode/ asynchronous mode Valid values ● wait (synchronous mode) ● nowait (asynchronous mode) If omitted: nowait setting will be used.	O	v2.3 or later
	destination destination of broadcasting or sending Valid values ● speaker (on a receiver) ● multicast1 (on a sender) … ● multicast20 (on a sender) ● sip (on Remote Mic)	Receiver: O Sender: M (If pattern_number is valid)	v2.3 or later
	target call target IP Address or SIP User ID when destination is “sip”. Ex: 192.168.14.2	O	v2.3 or later
	dtmf dtmf to send from Remote Mic Ex: “01” See chapter 7.	O	v2.3 or later

Output parameters

<command>	Command response	Available API Version
upload	status broadcasting state before issuing the command. (see 5 and 10.3.2)	v2.3 or later
	status_result operation status string as command result. (see 5 and 10.3.2) Exists only if param runmode equals to wait.	v2.3 or later

10.13 Category “multicast“

See 9.2.13 for command meaning

URL Syntax

```
http://<ipaddress>/api/v2/multicast/<command>?param1=XXX
```

Input parameters

<command>	Parameter	Mandatory Parameter	Available API Version
set_receive_channels	active enable/disable multicast receiving feature Valid values: "on" or "off" If omitted: keep setting	O	v2.3 or later
	enableXXX (XXX = 1 ～ 20) example: enable7 enable/disable multicast channel Valid values: "on" or "off" If omitted: keep setting	O	v2.3 or later
	addressXXX (XXX = 1 ～ 20) example: address7 receiving multicast address Valid values: class D IP address (example: 239.239.14.1) If omitted: keep setting	O	v2.3 or later

	portXXX (XXX = 1 ～ 20) example: port7 port number Valid values: 1024 ～ 65532 If omitted: keep setting	O	v2.3 or later
	nameXXX (XXX = 1 ～ 20) example: name7 Multicast Group Name Valid values: same as Multicast Screen If omitted: keep setting	O	v2.3 or later
get_receive_channels	channel channel number to be acquired Valid values: 1 to 20 If omitted: returns data for all channels	O	v2.3 or later

Output parameters

<command>	Command response	Available API Version
set_receive_channels	active enable/disable multicast receiving feature	v2.3 or later
	channels Channel information after changed Return as Object array with following keys. address: receiving channel channel: channel number enable: enable/disable name: Multicast Group Name port: port number	v2.3 or later
get_receive_channels	active enable/disable multicast receiving feature	v2.3 or later
	channels Channel information after changed Return as Object array with following keys If channel number is specified in input parameter, return only one channel. Otherwise return all channels.	v2.3 or later

set_receive_channels/ get_receive_channels commane response example

Return information for 20 channels under channels key (or information for the specified channel)


```
{
    "response": {
        "active": true,
        "channels": [
            {
                "address": "239.239.14.1",
                "channel": 1,
                "enable": true,
                "name": "Multicast 1",
                "port": 48000
            },
                ...
            {
                "address": "239.239.14.20",
                "channel": 20,
                "enable": false,
                "name": "Multicast 20",
                "port": 48038
            }
        ]
    },
    "result": true
}
```

11 Example

Example in actual use-case
See also chapter 7 for dtmf parameter.

11.1 Get supported command information

URL

http://192.168.14.1/api/v2/info/command

11.2 Get broadcast information

URL

http://192.168.14.1/api/v2/info/status

11.3 Play Pattern 10 for one hour without Interval

URL

http://192.168.14.1/api/v2/pattern/play?pattern_number=10&duration=3600&interval=0

11.4 Send Pattern 20 to Multicast 2

URL

http://192.168.14.1/api/v2/pattern/play?pattern_number=20&destination=multicast2

11.5 Stop broadcast or multicast sending for Pattern 1

URL

http://192.168.14.1/api/v2/pattern/stop?pattern_number=1

11.6 Stop broadcast or multicast sending for active Pattern without pattern number

URL

http://192.168.14.1/api/v2/pattern/stop

11.7 Start Patten 1 to SIP

URL

http://192.168.14.1/api/v2/pattern/play?pattern_number=1&destination=sip&target=192.168.1

11.8 Start microphone to SIP

URL

http://192.168.14.1/api/v2/microphone/play?destination=sip&target=192.168.14.3

11.9 Start multicast 2 from microphone

URL

http://192.168.14.1/api/v2/microphone/play?destination=multicast2

11.10 Enable multicast receiving feature

URL

http://192.168.14.1/api/v2/multicast/set_receive_channels?active=on

11.11 Set address of Receiving multicast 19 to 239.239.14.100

URL

http://192.168.14.1/api/v2/multicast/set_receive_channels?address19=239.239.14.100

11.12 Set multiple Receiving multicast channels

Set address of receiving Multicast 1 to 239.239.14.1
Set address of receiving Multicast 20 to 239.239.14.200 and the port to 50000
Enable Multicast 5

URL

http://192.168.14.1/api/v2/multicast/set_receive_channels?address1=239.239.14.1&address20=239.239.14.200&port20=50000&enable5=on

11.13 Upload Audio file (hoge.wav) to media 1 with same filename

Example command with curl (windows)

```
curl     ^
--digest    --user admin:guest                    ^
-X POST                                           ^
-H "Content-Type:application/octet-stream"        ^
--data-binary @hoge.wav                           ^
"http://192.168.14.238/api/v2/media/upload?media_number=1^&filename=hoge.wav"
```

Point

^ is line continuation character for windows batch file

-X POST is passed to use POST method

“application/octed-stream” is specified with Content-Type

Actual audio data is passed with --data-binary @filename

Uploaded media location is passed as Query string of URL.

Uploaded media filename is passed as Query string of URL.

11.14 Upload Audio file (test.mp3) to media 1 as あ.mp3

Example command with curl (windows batch file)

```
curl     ^
--digest --user admin:guest                       ^
-X POST                                           ^
-H "Content-Type:application/octet-stream"        ^
--data-binary @test.mp3                           ^
"http://192.168.14.238/api/v2/media/upload?media_number=19^&filename=%%E3%%81%%82.mp3"
```

Point

Uploaded media filename is passed as Query string of URL in UTF-8 encoded string.

“あ” is UTF-8 encoded as %E3%81%82

% is interpreted as special chars. Parameter is %%E3%%81%%82.

11.15 Upload audio file (test.mp3) to media 1 without filename and start Pattern 1

Example command with curl (windows batch file)

```
curl     ^
--digest    --user admin:guest                    ^
-X POST                                           ^
-H "Content-Type:application/octet-stream"        ^
--data-binary @test.mp3                           ^
"http://192.168.14.238/api/v2/media/upload?media_number=1^&pattern_number=1"
```

Point

Omit filename

Start Pattern 1 after upload

12 Reference Information

If you use Remote API on Windows command prompt using curl, you need to quote URLs with double quote(“)

This is because “&” is treated by Windows command prompt as special character.

Example of URL

```
http://192.168.14.1/api/v2/pattern/play?pattern_number=20&destination=multicast2
```

Example of curl command (it is displayed in two lines here. but run in single line command.)

```
curl --anyauth --user admin:guest
"http://192.168.14.1/api/v2/pattern/play?pattern_number=20&destination=multicast2"
```

Example of curl command (it is displayed in two lines here. but run in single line command.)
In the example, ignore proxy setting on the PC and send the command directly.

```
curl --noproxy “*” --anyauth --user admin:guest
"http://192.168.14.1/api/v2/pattern/play?pattern_number=20&destination=multicast2"
```

Example of curl command in https using Self Signed Certificate (Adding –k option)

```
curl –k --anyauth --user admin:guest
"https://192.168.14.1/api/v2/pattern/play?pattern_number=20&destination=multicast2"
```

See curl manual on how to use it.